

Project ID :

25-26J-369

1. Topic (12 words max)

Smart Mosquito Control System

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In recent years, Sri Lanka has witnessed an alarming increase in mosquito-borne diseases such as dengue and chikungunya, especially in densely populated urban regions. These illnesses are primarily transmitted by the *Aedes aegypti* mosquito, which thrives in tropical environments and breeds in stagnant water commonly found in domestic settings. Seasonal outbreaks continue to pose a severe burden on public health infrastructure, with thousands of reported cases annually, despite ongoing control efforts by local authorities.

Traditional mosquito control methods, including fogging and conventional traps, are often non-selective, inefficient, and lack the ability to provide real-time situational awareness. These approaches fail to distinguish between disease-carrying mosquitoes and harmless insect species, leading to ecological disruptions and suboptimal targeting. Moreover, most current systems operate without the integration of automated data collection or AI-based mosquito identification, making them reactive rather than preventative.

Another critical gap lies in the absence of centralized, data-driven platforms that link entomological surveillance with real-time patient case data. Health authorities often lack timely and localized information to take early action against disease outbreaks. Citizens, too, remain uninformed about risk levels in their surroundings, reducing the effectiveness of community-level preventive measures.

This research aims to address these gaps by developing a smart, AI-integrated IoT system for selective mosquito detection and real-time public health monitoring. The system will combine edge AI models for mosquito recognition (via image and sound analysis), an eco-friendly smart trap mechanism, and a cloud-connected dashboard that maps mosquito activity and outbreak hotspots. Public hospitals will input live dengue/chikungunya case data into a web/mobile app, which will also display cluster heatmaps to users, enabling both health officials and citizens to take proactive steps to reduce vector-borne disease transmission.

By integrating intelligent mosquito detection, IoT-based monitoring, and public health analytics, this project provides a scalable and actionable solution for urban mosquito control in developing countries like Sri Lanka.

1. World Health Organization (WHO). (2023). Dengue and severe dengue. Retrieved from: https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue
2. Ministry of Health – Epidemiology Unit. (2023). Dengue update Sri Lanka. Retrieved from: http://www.epid.gov.lk/web/index.php?option=com_content&view=article&id=171&lang=en
3. Harapan, H., Michie, A., Mudatsir, M., et al. (2019). Dengue: A review of current developments in detection and treatment. *Infectious Disorders - Drug Targets*, 19(2), 125–139.
4. Chen, T., et al. (2022). Smart mosquito trap with AI-based species identification and real-time monitoring. *IEEE Internet of Things Journal*, 9(4), 3254–3263.
5. Jayarathna, C., & Ekanayake, J. (2021). Review of dengue surveillance and vector control practices in Sri Lanka. *Sri Lankan Journal of Infectious Diseases*, 11(2), 67–74.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a smart AI-integrated IoT system designed to support selective mosquito control and real-time public health monitoring in urban Sri Lanka. It consists of four interconnected modules that collectively contribute to vector-borne disease prevention.

1. **AI-Based Identification:** Captured insects are analyzed using AI-based models that process images and wingbeat sounds. The system identifies whether the captured mosquito belongs to disease-spreading species like *Aedes aegypti*, while ignoring non-harmful species.
2. **Smart Trap Mechanism:** The system begins with an eco-friendly trap that attracts mosquitoes using a combination of UV light, CO₂ emission, and heat. Once mosquitoes are captured, they are held in a containment area for further identification.

3. IoT Sensor Network & Cloud Dashboard: Environmental data (e.g., temperature, humidity, CO₂) and mosquito detection logs are transmitted via IoT modules to a centralized cloud dashboard. The dashboard generates heatmaps of mosquito activity and environmental conditions in real time.

4. Health Data Integration & Alerts: A public-facing mobile/web app displays outbreak zones based on real-time hospital data input. This not only helps health officials coordinate targeted responses but also enables the public to take preventive actions when living in high-risk zones.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The proposed AI-integrated IoT system for smart mosquito control in urban Sri Lanka requires specialized expertise across multiple domains. AI and Machine Learning: Expertise in developing lightweight convolutional neural networks and audio signal processing is essential for real-time mosquito identification using image and wingbeat sound analysis. Knowledge of edge AI deployment on resource-constrained devices is critical. IoT and Sensor Networks: Proficiency in designing low-power sensor networks, integrating environmental sensors (e.g., temperature, humidity), and managing cloud-based data pipelines is needed for real-time mosquito activity monitoring. Embedded Systems and Hardware Design: Skills in prototyping eco-friendly smart traps with UV light, CO₂ emission, and electro-mechanical components are required, ensuring minimal power consumption and selective targeting. Web and Mobile Development: Expertise in full-stack development, data visualization, and API integration is necessary to create a mobile/web platform for outbreak heatmaps and health data integration.

Knowledge Requirements: Understanding mosquito behavior, dengue epidemiology, and Sri Lankan urban ecosystems is vital for tailoring the system. Familiarity with public health data protocols ensures seamless hospital data integration.

Data Requirements: The project needs access to open datasets like WINGBEATS for wingbeat sound profiles and localized Sri Lankan insect footage for AI training. Real-time environmental data (temperature, humidity) and hospital-reported dengue/chikungunya case data from Sri Lankan health authorities are crucial. Collaboration with the Ministry of Health or local hospitals will facilitate access to anonymized health data. Field-collected mosquito samples and environmental readings from urban areas will enhance model accuracy and system reliability.

8. Objectives and Novelty

Main Objective To design and implement an AI and IoT-integrated mosquito control system that selectively targets dengue-spreading mosquitoes and provides actionable data to improve public health responses in urban Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Pramod U.A.T IT22600202	Develop AI-based real-time mosquito identification	Collect localized insect data, train lightweight AI model with image/sound analysis	Lightweight edge deployable model tailored to Sri Lankan mosquito species
Wickrama Arachchi A.D.P.R IT22578600	Design eco-friendly smart trap mechanism	Prototype trap with UV light, CO2, and selective electro-mechanical system	Selective targeting of mosquitoes without harming non-target species
Rathnayaka R.M.G.M IT22568946	Build IoT sensor network and cloud monitoring system	Deploy sensors, develop real-time heatmap dashboard	Real-time mosquito density analytics via location-based heatmaps
Mithara M.D.N IT22604576	Create mobile/web health data integration platform	Integrate hospital data, visualize outbreak heatmaps	Dynamic outbreak heatmaps accessible to public and authorities for proactive action

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Pramod U.A.T IT22600202	Applies AI and machine learning specialization to develop a real-time mosquito identification model using image and sound analysis, aligning with expertise in data-driven systems.
Wickrama Arachchi A.D.P.R IT22578600	Utilizes embedded systems and hardware design specialization to prototype an eco-friendly smart trap with UV light and CO2 mechanisms, ensuring selective mosquito control.
Rathnayaka R.M.G.M IT22568946	Leverages IoT and networking specialization to design a sensor network and cloud-based dashboard for real-time mosquito density monitoring, enhancing data integration skills.
Mithara M.D.N IT22604576	Employs web and mobile development specialization to build a health data integration platform with outbreak heatmaps, showcasing expertise in user interface and data visualization.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Anjalie	Gamage	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.